



Fig. 7: (a) Sensor-based and (b) sensorless 3-phase BLDC, based on Atmel AVR microcontroller

variable-speed control methods are 'phase on/off time delay' and PWM control. The programmability of the microcontroller allows many features to be easily incorporated by changing the related firmware.

Applications galore

BLDC motors are replacing brush motors in numerous applications as BLDC motors offer many advantages including significant energy efficiency

improvements and better reliability. BLDC motors are typically smaller and lighter, which makes these ideal for applications where weight and space are important. The nonentity of brushes in a BLDC motor is perhaps its greatest advantage. The carbon brushes within a brush DC motor wear out rapidly, needing replacement. So these motors could end up costing you more. BLDC motors produce less noise, and are less prone to

sparking as these lack a commutator.

BLDC motors are now widely used in automotive, aerospace, military and medical applications. Gadgets that use BLDC motor include air-conditioners, computers, electric pumps, fans, printers, robots, electric bikes, food processors, blenders, vacuum cleaners, tooth-brushes, razors and coffee grinders to name a few. ●

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